Python with Database

```
python can be connected to both SQL and non SQL
```

MongoDB

```
noSQL database
stores data in a json-like document -> flexible
```

SQL vs NoSQL

SQL	NoSQL
Database	Database
Tables	Collections
Rows	Documents
Columns	Fields
Index	Index
Join	Embedding and Linking
Group by	Aggregation
Primary Key	_id field

Connect MongoDB to VSCode

```
1. go to mongodb website
2. sign up
3. select free plan
4. create a cluster with free region
5. go to network access and allow access from anywhere
6. add username and password
7. select python driver and copy the url link
8. connect your app with mongodb
9. install pymongo and dnspython (to use mongodb +srv://)
```

```python
import pymongo
MONGODB_URL = ""
client = pymongo.MongoClient(MONGODB_URL)
print(client.list_database.names())
# default mongodb databases
```

Create Database and Collections

```python
db = client.database_name
#OR
db = client["database_name"]
```

Insert Document

```
db.database_name.insert_one({"": "", "":""})
#OR
db.database_name.insert_many([])
```

Find Data

```
.find_one() # get the first occurence
.find_one({"_id": ObjectId("id")}) # specific id
.find() # all the occurence
.find({}, {"_id": 0, "name": 1}) #0 means not include, 1 means include
query = {
    "country": "Indonesia"
}
.find(query)
```

Query with Modifiers

```
query = {"age": {"$gt": 30}}
```

```
gt = greater than
lt = less than
gte = greater than or equal
lte = less than or equal
eq = equal
ne = not equal
in = in list
nin = not in list
```

Limiting Documents

```
db.students.find().limit(n)
```

Sort Documents

```
db.students.find().sort(age) #ascending
db.students.find().sort(age, -1) #descending
```

Update with Query

```
query = {"age": 120}
new_value = {"$set": {"age": 38}}
```

```
db.students.update_one(query, new_value)
#OR
db.students.update_many(query, new_value)
```

Delete Documents

```
db.students.delete_one(query)
db.students.delete_many(query)
db.students.delete_many() #delete all documents
```

Drop Collection

```
db.students.drop() #delete a collection
```